

$$2g - 2g(M + m)$$

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$$\begin{aligned} \text{efficiency} &= \frac{\text{useful output}}{\text{input}} \times 100\% \\ &= \frac{\text{electrical power}}{\frac{mgh}{t}} \times 100\% \\ &= \frac{100 \text{ MW}}{\frac{m}{t} gh} \times 100\% \\ &= \frac{100 \text{ MW}}{1.5 \times 10^5 (9.81)(140)} \times 100\% \\ &= 49\% \end{aligned}$$

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